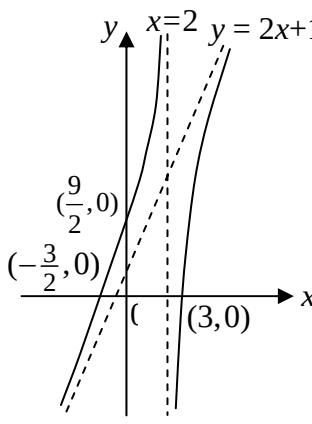
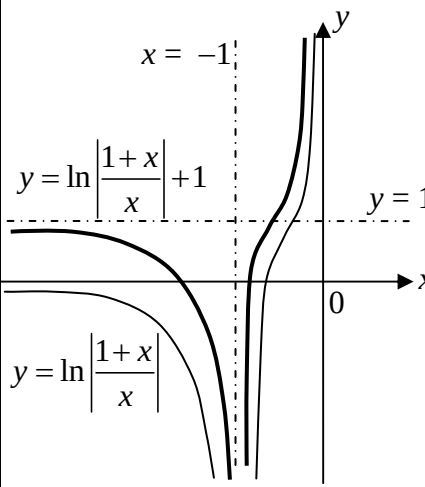
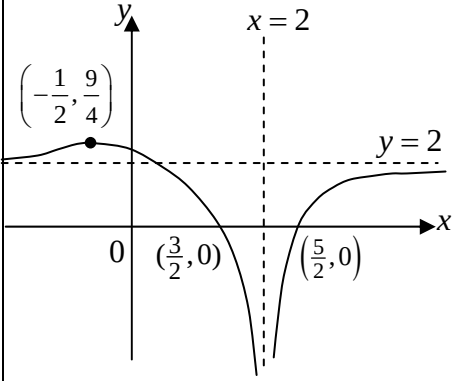
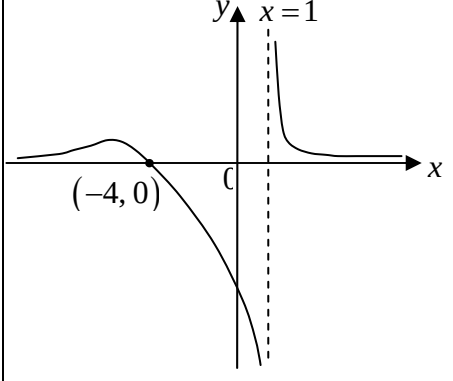
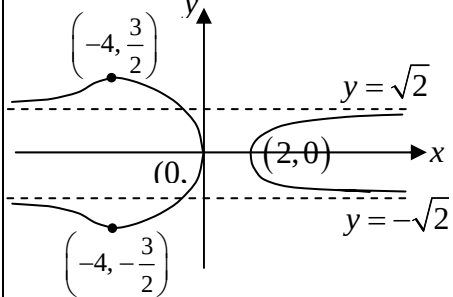


Annex B

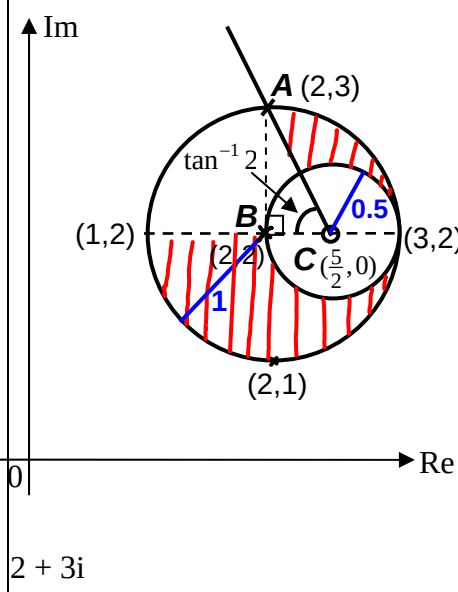
Cover Page for RI (JC) H2 Maths Prelim Examination Paper 1

Qn/No	Topic Set	Paper Number	Answers	Marks
1	Inequalities	9740/01	$-4.41 \leq x \leq 1.30$ or $x \geq 6.44$	4
2	Maclaurin's Series	9740/01	$y = 1 - \frac{1}{5}x - \frac{27}{250}x^2 + \dots$	5
3	Small Angle Approximation	9740/01	$a = 2000, b = -1000$	5
4	Techniques of Integration	9740/01	(a) $\frac{1}{3}\sec^3 x + c$ (b) $\frac{1}{6}\ln\left \frac{\sqrt{x}-3}{\sqrt{x}+3}\right + c$	6
5	Vectors	9740/01	(i) $\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$ (ii) $\begin{pmatrix} 6 \\ 4 \\ 6 \end{pmatrix}$ (iii) $b = 0$	6
6	Vectors	9740/01	(i) 80.3° (ii) 55.4°	6
7	Recurrence Relations	9740/01	(ii) $q = 0.1, u_{20} = 1000000$ Formulation of the recurrence relation did not take into account the physical constraints of the jar. (iii) The suggestion should not be taken up as the DE gives a continuous model as an approximation which only works when the population is large enough.	7
8	Mathematical Induction	9740/01	$(n+1)! - 6$	7
9	Applications of Integration	9740/01	$(-\frac{1}{\sqrt{2}}, \frac{1}{2})$ and $(\frac{1}{\sqrt{2}}, \frac{1}{2})$ 0.34 (2 d.p.), $\frac{5}{24}\pi$	8

10	Curve Sketching and Transformations	9740/01	$-2, a < -\frac{8}{5}$  Scale C parallel to the y-axis by a factor of $\frac{1}{3}$, then translate 5 units in the positive y-direction Alternatively, Translate C 15 units in the positive y-direction, then scale parallel to the y-axis by a factor of $\frac{1}{3}$.	9
11	Differential Equations	9740/01	(i) $y = \ln \left \frac{1+x}{x} \right + C$ (ii) $y = \ln \left \frac{1+x}{x} \right$ 	8
12	AP and GP	9740/01	(a)(ii) $\frac{1}{2}n(n^2+1), 59$ (b) $2a$	9

13	Curve Sketching	9740/01	(i) $y = f(2x - 3)$  (ii) $y = f'(x)$  (iii) $y^2 = f(x)$ 	9
14	Complex Numbers	9740/01	(a)(ii) $z - (1 + i) = \sqrt{2}$ $n = 4m + 2, m \in \mathbf{Z}, m \geq 0$ (b) $1 + i + \sqrt{2}e^{\frac{2k\pi i}{5}}, k = 0, \pm 1, \pm 2$ $(x - 1)^2 + (y - 1)^2 = 2$	11

Cover Page for RI (JC) H2 Maths Prelim Examination Paper 2

Qn/No	Topic Set	Paper Number	Answers	Marks
	Pure Maths			
1	Complex Numbers	9740/02		8
2	Functions and Recurrence	9740/02	(i) $ff : x \mapsto \sqrt{\sqrt{x+2}+2}, x \geq -2$ Range of ff is $[\sqrt{2}, \infty)$ (iii) $v = 2$ (iv) $\{2\}$	9
3	Applications of Differentiation	9740/02	$2\pi \text{ cm}^3 \text{ s}^{-1}$	11
4	Vectors	9740/02	(i) $\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix} = -5$ (ii) $\vec{OB} = \frac{1}{41} \begin{pmatrix} -14 \\ 75 \\ -42 \end{pmatrix}, \frac{7\sqrt{41}}{41} \text{ units}$ (iii) $4\sqrt{\frac{10}{41}} \text{ units}$ (iv) $\frac{14\sqrt{10}}{41} \text{ units}^2$	12

	Statistics			
5	P & C	9740/02	144, 11520	5
6	Prob	9740/02	(a) (i) 0.4 (ii) 0.5 (b) (i) 0.166 (ii) 0.119	8
7	Binomial, Poisson and Normal	9740/02	(a) 0.145 (b)(i) 0.677 (ii) 0.333	10
8	Normal	9740/02	(i) $k = 21.6$ (ii) $E(Y) = 12$, $\text{Var}(Y) = 32$; 0.533	9
9	Sampling	9740/02	(iii) 5.64, 7.75 (iv) 0.180	8
10	Hypothesis Testing	9740/02	(ii) z-test is the correct test because X follows a normal distribution and $\text{Var}(X)$ is known (iii) not possible	8
11	Correlation and Regression	9740/02	(i) 0.950 (ii) (4.7, 46.9) (iii) 117.92608 marks Estimate is not reliable as $x = 12$ is not within the domain of the given data set. (iv) $y = 101 - \frac{198}{x}$ 85.0 marks The scatter plot of y against $\frac{1}{x}$ below suggests a linear relation between of y and $\frac{1}{x}$, whereas that of y against x in (ii) does not.	12